AI PRODUCT MANAGEMENT CASE STUDY · FICTIONAL TARGETS AND REAL-FILING BACKTESTS · NOVEMBER 2026

Project Falcon

Multi-agent M&A due diligence, built in Claude Code without application code

An AI diligence team that researches an acquisition target across SEC filings, news, court records and financials, connects what no single source shows, and drafts an Investment Committee report, with people approving each stage.

12	97.4%	18 of 18	38 → 13	0
blind graded runs on 6 fictional targets	Critical and High recall, runs 1 to 3 average	issues found on each of two held-out targets	report pages, runs 4 to 6	unsupported findings in any report

The problem

Diligence takes teams of associates weeks of reading filings, news, dockets and data-room financials, and the worst issues hide between sources. Autonomous agents could help, but they are the hardest AI pattern to control, so high-stakes work needs human checkpoints.

What I built

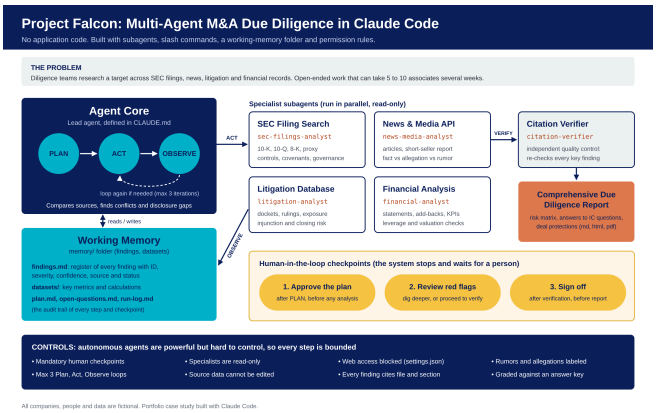
- **Agent Core:** a lead agent (`CLAUDE.md`) running a bounded Plan, Act, Observe loop (at most three iterations).
- **Four specialists in parallel:** SEC filings, news, litigation and financials. One source each, read-only.
- **Independent citation verifier:** re-opens every cited source and recomputes the numbers behind each Critical and High finding.
- **Working memory as plain files:** plan, findings register, conflicts, run log and every unedited agent report. Fully auditable.
- **Three human checkpoints:** approve the plan, review the red flags, sign off.
- **Testable guardrails:** no web access, protected sources, a prompt-injection rule, approval to open the answer key, 69 automated checks.

Key decisions

- **Specialists by source, not topic,** so connecting sources is the lead agent's job. The Critical issues only appear there.
- **Separation of duties:** analysts cannot change anything; a separate verifier checks them, like a deal-team reviewer.
- **Measured, not assumed:** 18 planted issues, 4 distractors and an answer key the analysts and verifiers never see.
- **Planned like a product:** specs and a phased plan with tests for each phase, agreed before building.

Real filings: two backtests

Two backtests ran on a real company's SEC filings as of a cutoff date (2018 and 2014), with a hypothetical deal and a key of warning signs the later public record showed mattered. **Limits first:** one company each, picked by a session that knew the outcome; no distractors, so false alarms are untested; the companies and reports stay private. Found: 10 of 10 signs, then 5 of 8 plus 3 in part (81.3% scored, 62.5% on full finds alone). Both: 0 unsupported findings, 0 hindsight leaks.



Architecture: Agent Core, four specialists, working memory, verifier and human checkpoints.

What the system concluded

All six Falcon runs: do not proceed at $24.00 per share. The committee decides.

1. The valuation used guidance the target had already replaced: about 18.3x on current guidance, not the 11.5x in the deal model.
2. A patent case turned against the target in July, with an injunction sought on robots behind about 42% of revenue. The next quarterly filing left the ruling out.
3. A customer worth 31% of revenue can leave after a change of control, and is building its own robots.

What I learned

- The verifier earned its place: 17 of 61 findings needed corrected wording before the report (one claimed 18 months of negative free cash flow; it was 12).
- Three runs, not one: the same misses repeated, so they were design gaps, not luck. I fixed them, and run 4 found all 18.
- Then quality and generalization: reports fell from 38 to 13 pages. On four held-out targets it found 96% to 100% of Critical and High issues and invented nothing; exact severity was 77.8%, 88.9%, 72.2% on Wren and 72.2% on Avocet, where three of the four test items for Wren's rules passed.

How blind was the test? Each graded run started in a fresh session that never saw the answer key or an earlier run. Heron's, Wren's and Avocet's builders never saw the instructions; Plover's could. A different session graded Wren and Avocet.

Cost and time: agents versus two analysts

One real-company backtest, from start to finished report, if the same agents ran on the paid API, set against two analysts doing the same first pass by hand.

	This system	Two analysts (illustrative)
Cost	**US$68 to US$70**	**US$16,000 to US$48,000**
Elapsed time	**1.1 to 1.4 hours**	**about 2 weeks**
Working time	**about 1 hour of agent work, plus 4 to 22 minutes of a person answering three checkpoints**	**160 person-hours**
Basis	Measured from the two backtests' session logs: 108 to 126 million tokens at Claude Opus 5.5 API list rates (96% were cache reads, which cost a twentieth of normal input), and timestamps for the time.	My assumptions, not measured: two analysts for two working weeks at US$100 to US$300 an hour.

What this means: the first pass costs under 1% as much and finishes in about an hour instead of two weeks. It covers only the first-pass reading and cross-checking. People still decide at all three checkpoints, and a reviewer reading the report is not in these figures. The project ran on a Claude subscription, so the cost is the API price of the same work, not what I paid.

In the repository: README (with a demo video under 2 minutes and both backtests), `runs/` (graded runs with scorecards), `examples/sample-run/` and `Project Memory/` (specs and plan).

Skills: AI product management (specs, phased plan, success metrics, evaluation, iteration from data), agentic workflow design and M&A domain depth. The fictional targets' companies, people and figures are invented; the backtests' real companies stay unnamed. Not investment advice.